

BrainForge

Complete Handbook

User Guide • Infrastructure • Deployment • Recovery

From GitHub API to Cloudflare Deployment

Everything you need to rebuild BrainForge from scratch

CRITICAL RESOURCES

Live App: brainforge-7xn.pages.dev

GitHub: [richardbrownmiami-commits/devforge-ai](https://github.com/richardbrownmiami-commits/devforge-ai)

Worker: brainforge-api.richard-brown-miami.workers.dev

Telegram Bot: [@araislivingbot](https://t.me/araislivingbot)

D1 Database: brainforge-db

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BrainForge is a self-hosted, AI-powered web app builder. It lets you describe app ideas in chat, have AI generate HTML/CSS/JS code, preview it live, and deploy to Cloudflare Pages. It is completely independent of any single platform.

Architecture Overview

Layer	Technology	Details
Frontend	React + TypeScript + Tailwind CSS	Vite build, deployed on Cloudflare Pages
Backend API	Cloudflare Workers (JavaScript)	brainforge-api.richard-brown-miami.workers.dev
Database	Cloudflare D1 (SQLite)	brainforge-db
Code Storage	GitHub Repository	richardbrownmiami-commits/devforge-ai
Local Storage	IndexedDB + localStorage	Chat history, settings, snapshots
AI Providers	Gemini 2.0 Flash + OpenRouter	Free models, auto-rotation on rate limits
Telegram Bot	Cloudflare Worker	@araislivingbot

Key URLs

Service	URL
Live App	https://brainforge-7xn.pages.dev
API Worker	https://brainforge-api.richard-brown-miami.workers.dev
GitHub Repo	https://github.com/richardbrownmiami-commits/devforge-ai
Telegram Bot	https://t.me/araislivingbot
Cloudflare Dashboard	https://dash.cloudflare.com

Platform	URL	Purpose
GitHub	https://github.com	Code storage + CI/CD + session files
Cloudflare	https://cloudflare.com	Workers + D1 database + Pages hosting
OpenRouter	https://openrouter.ai	Free AI models (Llama, Gemma, Mistral)
Google AI Studio	https://aistudio.google.com	Gemini 2.0 Flash - 1500 req/day free
Telegram	https://telegram.org	For the Ara bot
BotFather	t.me/BotFather	To create the Telegram bot

API Keys You Will Need

- GitHub Personal Access Tokens (repo + contents permissions)
- Cloudflare API

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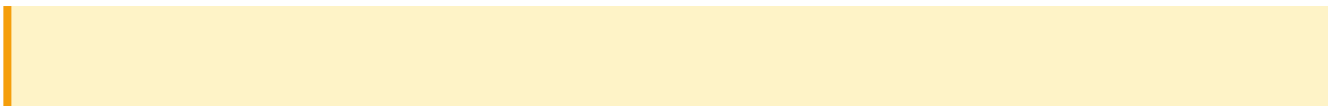
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Key (free registration)

- Google Gemini API Key (from AI Studio, free)

- Telegram Bot Token (from BotFather)

NOTE: Save all API keys in a secure password manager immediately after creating them. Never commit them to public repositories.



Go to <https://github.com/signup> and create a free account.

Step 2: Create new repository

- Click the + button to create a new repository (to get started with GitHub, you can create a new repository)
- Repository name: `dev-for-generative-ai` (or any name you want)

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- Set to Private (recommended to protect API keys)

- Check "Add a README file"

- Click "Create repository"

or
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Step 2: Create Personal Access Token (Classic)

Step 1: Go to token settings

Navigate to: GitHub > Settings > Developer Settings > Personal Access Tokens > Tokens (classic)

Step 3: Configure the token (classic)

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Step 4: Copy and save the token

The token starts with ghp_... Copy it immediately -- you cannot see it again.

Step 3: Add Token to BrainForge

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- Click GitHub & Deploy
- Paste your token in the GitHub Token field
- Enter your repository as: username

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Token format: ghp_XXXXXXXXXXXXXXXXXXXX

Repo format: username/repo-name

Example: richardbrownmiami-commits/devforge-ai

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Go to <https://dash.cloudflare.com/sign-up> and create a free account. No credit card required for Workers/D1/Pages free tier.

Step 2: Get your Account ID

- Go to <https://dash.cloudflare.com>
- Click any domain OR go to Workers & Pages
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Step 2: Create D1 Database

Step 1: Navigate to D1

In Cloudflare dashboard: Workers & Pages > D1 SQL Database > Create database

Step 2: Create the database

- Database name: brainforge-db
- Location: Auto (nearest to your user)
- Click


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CREATE TABLE IF NOT EXISTS rules (  
  scope TEXT PRIMARY KEY,  
  content TEXT,  
  updated_at INTEGER  
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Step 3: Create Cloudflare API Token

Step 3: Copy the Database ID

Cloudflare Dashboard > My Profile (top right) > API Tokens > Create Token

Step 2: Use "Edit Cloudflare Workers" template OR create custom token

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- Account Resources: Include > Your account

- Click Continue to Summary > Create Token

Step 3: Copy the token

Starts with: a Cloudflare token string. Save it immediately.

```
npm install --save wrangler
main = "src/index.js"
#verify installation "2024-01-01"
npm pkg set compat-version=1.0
```

```
[[{"databases"}]]
#Authenticate with Cloudflare
wrangler login
database_name = "brainforge-db"
database_id = "YOUR-D1-DATABASE-ID-HERE"
```

Wrangler is the CLI tool for deploying Cloudflare Workers and Pages.

Step 3: Worker Code Structure

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- GET/POST /api/rules - All rules files

- POST /api/sync - Sync all data to D1

- GET /api/stats - Data as seen

```
export CLOUDFLARE_API_TOKEN=your_token_here
export CLOUDFLARE_ACCOUNT_ID=your_account_id_here
```

```
# Add secrets that the Worker can read securely
# Deploy the worker
wrangler secret put GITHUB_TOKEN
or wrangler deploy
# (prompts you to paste the value)
```

```
# Expected output:
wrangler secret put OPENROUTER_KEY
e# Uploaded brainforge-api (x.xx sec)
```

NOTE: Worker free tier includes 100,000 requests/day and 10ms CPU time per request. More than enough for personal use.

```
# Published brainforge-api
# URL: https://brainforge-api.your-subdomain.workers.dev
wrangler secret list
```

```
S C O N F I G U R A T I O N F O R B R A I N F O R G E - A P I
```

```
# Deploy to Cloudflare Pages
CLOUDFLARE_API_TOKEN=your_token CLOUDFLARE_ACCOUNT_ID=your_account_id npx wrangler
pages deploy --project-name brainforge
# First time: creates the project automatically
# Subsequent times: creates a new deployment
# Preview URL: https://brainforge-7xn.pages.dev (or your chosen name)
```

Method B: Connect GitHub Repository
If this is the first time using Method B, please read the section from your project root:

Step 1: Go To Cloudflare Pages

Dashboard > Workers & Pages > Create application > Pages > Connect to Git
Build Command: cd src/Frontend && npm install && npx vite build

Authorize Cloudflare to access your GitHub repositories

NOTE: The Cloudflare project name must be consistent. Your current project name is: brainforge (not brainforge-7xn).

The URL brainforge-7xn.pages.dev is the assigned subdomain.

Click Save and Deploy. Every push to main branch auto-deploys.

```
# Set bot secrets
wrangler secret put TELEGRAM_BOT_TOKEN
wrangler secret put OPENROUTER_KEY
/newbot
# Enter bot name: Ara - BrainForge AI
# Enter username: araislivingbot (or your choice, must end in 'bot')
cd telegram-bot
# Botfather gives you a token like: 123456789:ABCdef...
Search for @BotFather and start a chat
This is your TELEGRAM_BOT_TOKEN. Save it securely.
# Register webhook
curl https://api.telegram.org/bot<TOKEN>/setWebhook --d url=https://your-bot-worker.workers.dev/webhook
```

Step 2: Deploy Bot as Cloudflare Worker

Step 3: Configure Bot Behavior

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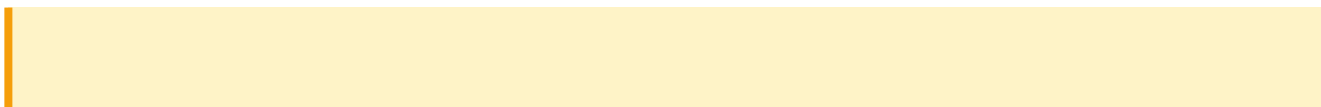
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NOTE: Current bot uses deepseek/deepseek-r1:free as primary model with meta-llama/llama-3.3-70b-instruct:free as fallback. Both are free on OpenRouter.



Provider	Models	Notes
Auto (Recommended)	Tries Gemini first, then OpenRouter	Best reliability
Google Gemini	gemini-2.0-flash + lite	1500 req/day free, fastest
OpenRouter	Llama, Gemma, Mistral, Qwen	100+ free models

Chat Screen

- Full-screen chat interface
- type instructions to build apps
- Preview but not to n (copyright) -- open

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Preview Screen

- Full-screen overlay showing your urgent reminder at the app
- HTML / CSS / JS tabs -- view and copy

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Settings Hub (7 sections)

Section	Purpose
API Keys	Add OpenRouter and Gemini API keys
AI Settings	Choose provider (Auto/OpenRouter/Gemini) + model
Termux Brain	Connect your Android phone as AI backend server
GitHub & Deploy	GitHub token + repo for saving and deploying apps
Master AI	AI that modifies BrainForge itself -- push changes to GitHub
AI Files	View and edit memory/rules files for each project
Database	View project files, AI memories, storage meter, export/import

Projects Page

- All your projects shown as cards
- Click a card to open the editor
- New Project button -- create with template

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Dark/Light Theme

Toggle in the sidebar -- Moon icon for dark mode, Sun icon for light mode. Preference is saved automatically.

Version History

BrainForge auto-saves a snapshot before every AI response. Click the Clock icon in the editor header to see the last 10 snapshots and restore any of them.

Auto Error Fix

If your generated app has a JavaScript error, BrainForge automatically detects it and asks the AI to fix it (up to 3 attempts). You will see the fix happening automatically.

```
sessions/{projectName}/session.json
```

Example:

```
sessions/my-quiz-app/session.json
```

Every time you chat with an AI in a project, BrainForge automatically pushes a session file to your GitHub repository. When you open a project with empty local storage, BrainForge automatically pulls from this file and restores

Manual Session Restore (if auto-restore fails)

Step 1: Go to your GitHub repo

```
https://github.com/YOUR_USERNAME/devforge-ai/tree/main/sessions
```

Step 2: Download the session file from the JSON file

Step 3: Use Import Backup to restore Database

Full Reset & Rebuild Procedure

```
git clone https://github.com/YOUR_USERNAME/devforge-ai.git
```

Step 4: Once BrainForge goes offline, follow these steps to rebuild:

Step 1: Open the Setup Pages

Step 5: Add API keys in Settings

Open the deployed app and re-enter your OpenRouter, Gemini, and GitHub tokens.

Step 6: Restore session data

NOTE: Always export your data from Settings > Database > Export All Data as JSON and save it locally as a backup. Use Settings > Database > Import Backup with your exported JSON file.

BrainForge AI Troubleshooting

Problem	Solution
AI not responding	Check API key in Settings > API Keys. Try switching to Auto provider
Rate limit error	Auto mode switches providers. Wait 60 seconds or add both Gemini + OpenRouter keys
Preview shows error	Click Refresh in preview. AI will auto-fix JS errors up to 3 times
Settings rolling back	Workspace resets on Caffeine session end. This is normal data is in localStorage
Push to GitHub fails	Check GitHub token has repo permissions. Token might be expired
Worker not found	Redeploy worker: cd worker && wrangler deploy
Deploy button fails	Check GitHub token and repo in Settings > GitHub & Deploy.
Telegram bot silent	Check webhook: curl https://api.telegram.org/bot<TOKEN>/getWebhookInfo

```
# test worker is running
curl https://brainforge-api.richard-brown-miami.workers.dev/api/status
# test response:
{"tables": [{"name": "messages", "total_rows": 1}]}
wrangler d1 execute brainforge-db --command="SELECT * FROM projects LIMIT 5"
# Verify:
# Reset all data (CAUTION)
wrangler d1 execute brainforge-db --command="DELETE FROM messages"
```

Re-registering Telegram Webhook

Checking Worker Health

Cloudflare D1 Queries

File	Purpose
src/frontend/src/pages/EditorPage.tsx	Chat + Preview screen
src/frontend/src/components/ChatPanel.tsx	Chat UI component
src/frontend/src/components/PreviewPanel.tsx	Preview iframe
src/frontend/src/pages/SettingsPage.tsx	Settings hub (7 buttons)
src/frontend/src/hooks/useAIChat.ts	AI provider + model logic
src/frontend/src/constants/models.ts	Available AI models
src/frontend/src/utlis/storage.ts	IndexedDB + GitHub session
worker/src/index.js	Cloudflare Worker API
worker/wrangler.toml	Worker config (D1 binding)
telegram-bot/ara-behavior.js	Bot personality config
memories/master-ai-memory.md	Master AI context
rules/master-ai-rules.md	Master AI rules
sessions/{name}/session.json	Auto-saved chat sessions

localStorage Keys (BrainForge data) Essential Commands

Key	Contains
bf_settings	API keys, provider, GitHub token, theme
bf_projects	Array of all project names + metadata
bf_chat_{name}	Chat message history for a project
bf_snapshots_{name}	Version history snapshots
bf_deploy_url_{name}	Live URL after deployment
bf_ai_file_{key}	AI memory/rules files
bf_starter_{name}	Template starter prompt
bf_theme	dark or light

Free Tier Limits Summary

Service	Free Tier Limits
GitHub	Unlimited public repos, 500MB storage, 2000 Actions min/month
Cloudflare Workers	100,000 requests/day, 10ms CPU per request
Cloudflare D1	5M reads/day, 100K writes/day, 500MB storage
Cloudflare Pages	Unlimited sites, 500 builds/month, 100GB bandwidth
OpenRouter Free	~10-50 req/min per model, shared quota
Google Gemini	1500 req/day, 15 req/min (gemini-2.0-flash)
Telegram	Unlimited messages, no rate limits for personal bots

You have everything you need.

BrainForge is fully self-hosted and independent.

Even if Caffeine disappears tomorrow, you can rebuild BrainForge in under 30 minutes using this guide.

[GitHub](#) • [Cloudflare](#) • [Your AI Keys](#) • [This Handbook](#)